Sets 6

Wednesday, 20 March 2024 15:29

① $t_0 = 0$ $\quad t_{n+1} = t_n + 2n + 3$

a) $t_1 \leftrightarrow t_{0+1} = t_0 + 2.0 + 3 = 0+0+3$

$t_2 \leftrightarrow t_{1+1} = t_1 + 2.1 + 3 = 3+2+3 = 8$

$t_3 = t_{2+1} = t_2 + 2.2 + 3 = 8+4+3 = 15$

$t_4 = t_{3+1} = t_3 + 2.3 + 3 = 24$

b) $\forall n \geq 0 : t_n = n(n+2)$

Proof – base case $\quad n = 0$

$\star\ t_0 = 0$

$\star\ t_0 = 0.(0+2) = 0.2 = 0$

Hence, $t_n = n(n+2)$ is TRUE for $n=0$

Proof – Inductive Step

Let $m \geq 0$ Assume

$t_m = m(m+2) \longleftarrow$ Induction Hypothesis

$t_{m+1} = t_m + 2m + 3$

$t_{m+1} = m(m+2) + 2m + 3 = (m+1)(m+3)$

$m^2 + 4m + 3 \quad = m^2 + 4m + 3$

Hence, IH ($t_m = m(m+2)$) implies $t_{m+1} = (m+1)((m+1)+2)$ for all $m \geq 0$

②

a) $t_{1+1} = 2t_1 - 2.1 + 3$

2.2

$= 4 - 2 + 3 = 5$

$t_{2+1} = 2.t_2 - 2.2 + 3$

$2.5 - 4 + 3 = 9$

$t_{3+1} = 2.t_3 - 2.3 + 3$

$2.9 - 6 + 3 = 15$

$t_{4+1} = 2t_4 - 2.4 + 3$

$2.15 - 8 + 3 = 25$

b) $t_n = 2^{n-1} + 2n - 1$

$t_1 = 2^0 + 2 - 1 = 2$ base case ✓

Let assume $m \geq 1$

$t_m = 2^{m-1} + 2m - 1$

$t_{m+1} = 2t_m - 2m + 3$

$= 2(2^{m-1} + 2m - 1) - 2m + 3$

$= 2^m + 4m - 2 - 2m + 3$

$t_{m+1} = 2^m + 2m + 1$

$t_{m+1} = 2^{(m+1)-1} + 2(m+1) - 1$

$= 2^m + 2m + 2 - 1$

$= 2^m + 2m + 1$

same

Hence $[t_m = 2^{m-1} + 2m - 1]$ implies $t_{m+1} = 2^{(m+1)-1} + 2(m+1) - 1$ for all $m \geq 1$

③ $t_1 = 1 \quad t_{n+1} = n + 3n + 1$

a) $t_2 \leftrightarrow t_{1+1} = 3.1 + 1 = 1+3+1 = 5$

$t_3 \leftrightarrow t_{2+1} = t_2 + 3.2 + 1 = 5+6+1 = 12$

$t_4 \leftrightarrow t_{3+1} = t_3 + 3.3 + 1 = 12+9+1 = 22$

$t_5 \leftrightarrow t_{4+1} = t_4 + 3.4 + 1 = 22+12+1 = 35$

b) $\forall n \geq 1 \quad t_n = \frac{1}{2}n(3n-1)$

Proof base case $\quad n=1$

$t_n = 1$

$t_n = \frac{1}{2}1(3.1-1) = \frac{1}{2}.2 = 1$

Hence $t_n = \frac{1}{2}n(3n-1)$ is TRUE for $n=1$

Proof Inductive step

Let $m \geq 1$ assu $t_m = \frac{1}{2}m(3m-1)$ IH

$\rightarrow t_{m+1} = t_m + 3m + 1 = \frac{1}{2}m(3m-1) + 3m + 1 = \frac{3m^2}{2} + \frac{5m}{2} + 1$

$\rightarrow t_{m+1} = \frac{1}{2}(3m^2 + 5m + 2)$

④ **a)** $n \geq 1 \quad \sum_{k=1}^{n} \frac{1}{k(k+1)} = \frac{n}{n(n+1)}$ $\qquad \frac{1}{1.2} + \frac{1}{2.3} \cdots + \frac{1}{n(n+1)} = \frac{n}{n+1}$

Proof – Base case : $n=1$

▶ $\sum_{k=1}^{1} = \frac{1}{k(k+1)} = \frac{1}{1(1+1)} = \frac{1}{1.2} = \frac{1}{2}$

▶ $\frac{n}{n+1} = \frac{1}{1+1} = \frac{1}{2}$

Hence, $\sum_{k=1}^{n} = \frac{1}{k(k+1)} = \frac{n}{n(n+1)}$ is TRUE for $n=1$

Proof – Inductive Case

Let $m \geq 1$. Assume $\sum_{k=1}^{m} \frac{1}{k(k+1)} = \frac{m}{m+1}$ ⒾⒽ

▶ $\sum_{k=1}^{m+1} \frac{1}{k(k+1)} = \sum_{k=1}^{m} \frac{1}{k(k+1)} + \frac{1}{(m+1)((m+1)+1)} \overset{IH}{=} \frac{m}{m+1} + \frac{1}{(m+1)(m+2)} = \frac{m+1}{m+2}$

▶ $\frac{m+1}{(m+1)+1} = \frac{m+1}{m+2}$

Hence, IH $\left[\sum_{k=1}^{m} \frac{1}{k(k+1)} = \frac{m}{m+1}\right]$ implies $\sum_{k=1}^{m+1} \frac{k}{k(k+1)} = \frac{m+1}{(m+1)+1}$ for all $m \geq 1$

b) $\forall n \geq 1 \quad \sum_{k=1}^{n} (k+1)2^k = n.2^{n+1}$ $\qquad 2.2^1 + 3.2^2 + 4.2^3 \ldots + (n+1).2^n = n.2^{n+1}$

Proof base case: $n=1$

▶ $\sum_{k=1}^{1} (k+1)2^k = (1+1).2^1 = 2.2 = 4$

▶ $n.2^{n+1} = 1.2^{n+1} = 1.2^2 = 1.4 = 4$

Hence $\sum_{k=1}^{n} (k+1)2^k = n.2^{n+1}$ is TRUE for $n=1$

Proof Inductive Step

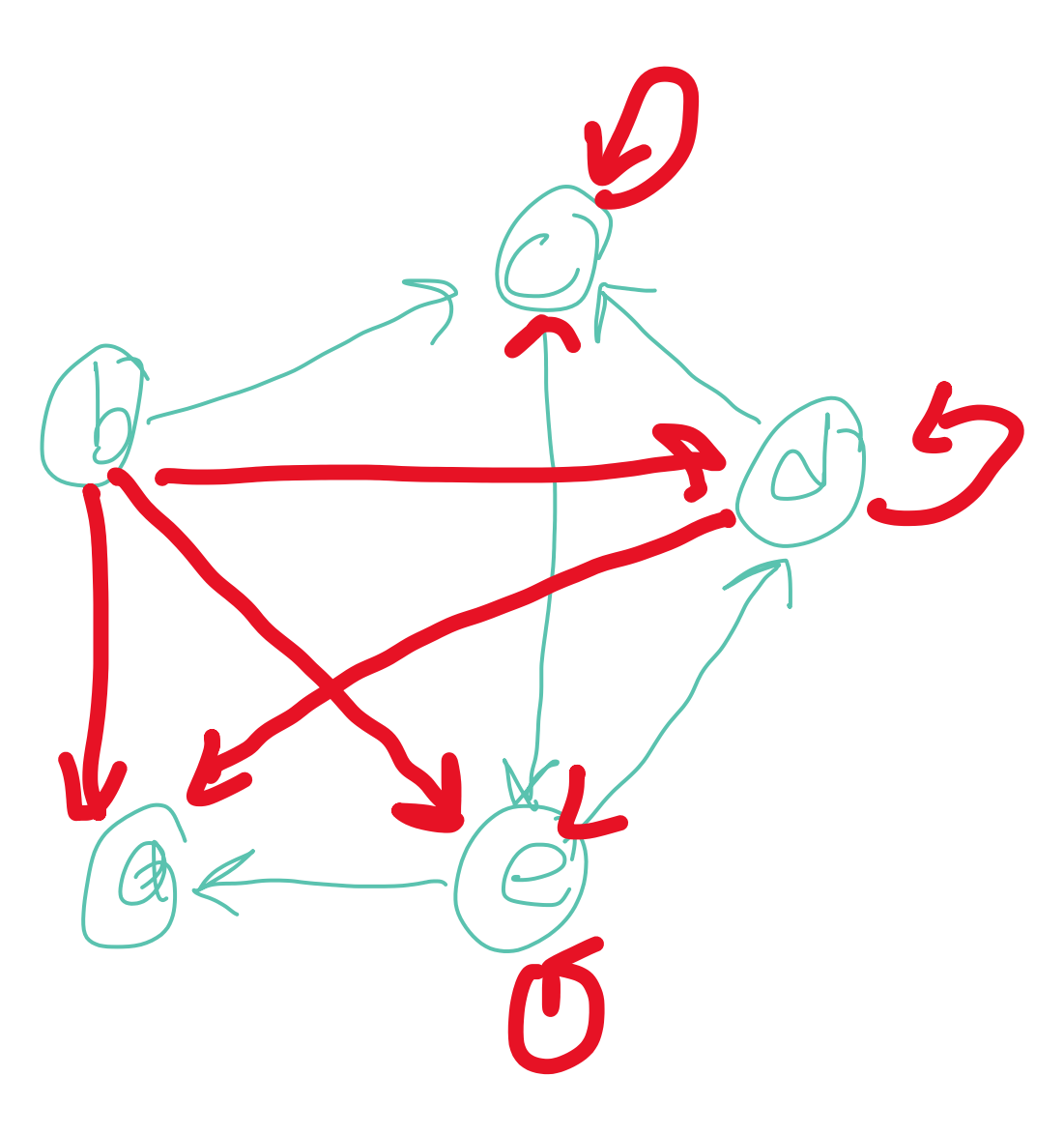
Let $m \geq 1$. Assume $\sum_{k=1}^{m} (k+1).2^k = m.2^{m+1}$ ⒾⒽ

▶ $\sum_{k=1}^{m+1} (k+1).2^k + ((m+1)+1).2^{m+1} \overset{IH}{=} m.2^{m+1} + ((m+1)+1).2^{m+1} = (m+1).2^{m+2}$

▶ $(m+1)2^{(m+1)+1} = (m+1)2^{m+2}$

Hence, IH . . . ✓

6



7) a) $xRy \iff x+1=y$

Transitive closure

$< = \{ \langle n,m \rangle \in \mathbb{N}^2 : n<m \}$

b)

$0 \longrightarrow 1$

$\searrow 10 \longrightarrow 11$

$\searrow 110 \longrightarrow 111$

$\searrow 110$

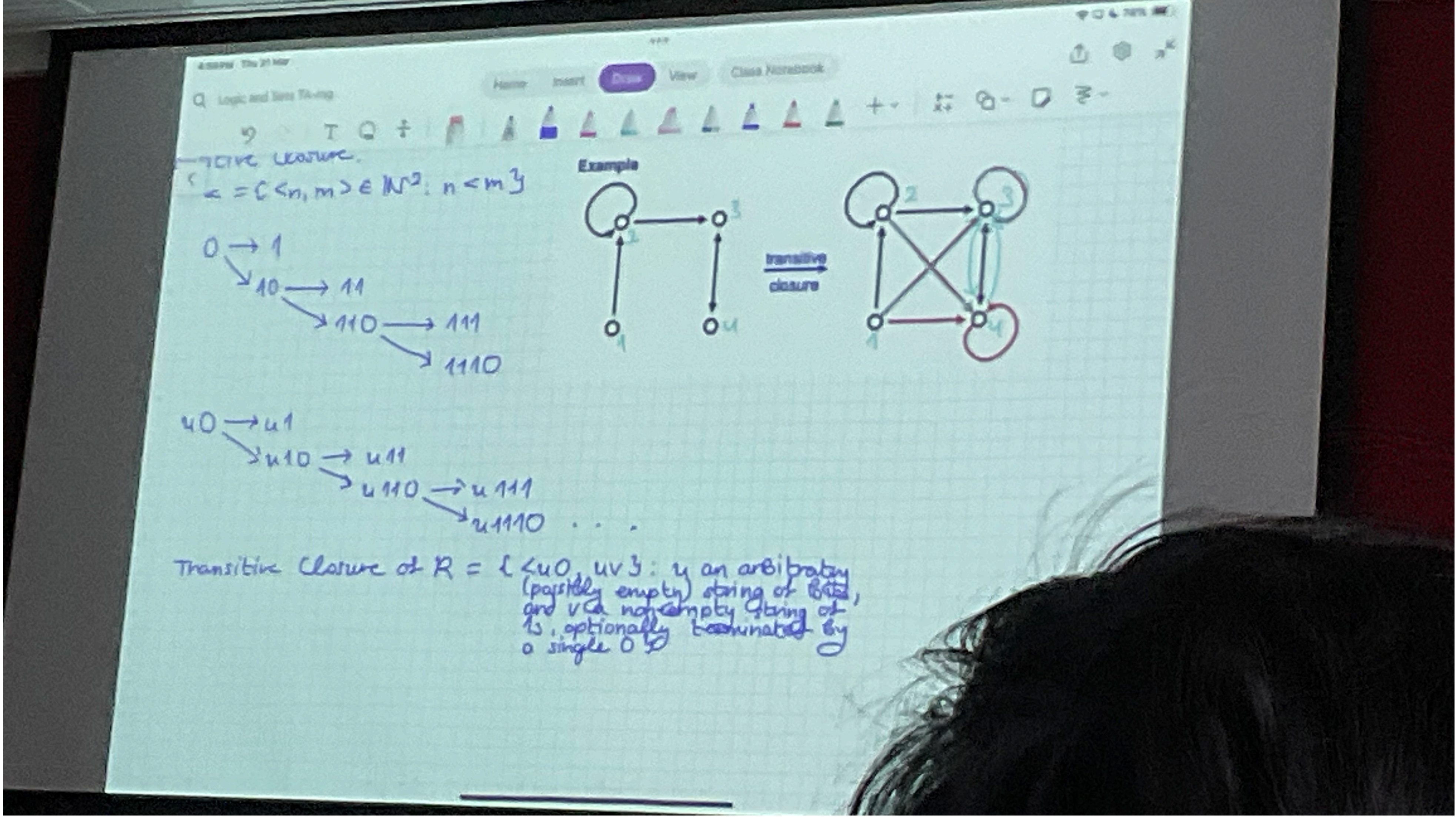
$u0 \longrightarrow u1$

$\searrow u10 \rightarrow u11$

$\searrow u110 \rightarrow u111$

$\searrow u1110$

Closure of $R = \{ \langle u0 \quad uv \rangle : u$ an



—ive closure.

$< = \{ \langle n,m \rangle \in \mathbb{N}^2 : n<m \}$

$0 \rightarrow 1$

$\searrow 10 \longrightarrow 11$

$\searrow 110 \longrightarrow 111$

$\searrow 1110$

$u0 \rightarrow u1$

$\searrow u10 \rightarrow u11$

$\searrow u110 \rightarrow u111$

$\searrow u1110$ . . .

Transitive Closure of $R = \{ \langle u0, uv \rangle$: u an arbitrary (possibly empty) string of bits, and v a non-empty string of 1s, optionally terminated by a single 0 $\}$